山东大学 计算机科学与技术 学院

计算机网络 课程实验报告

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实验题目：计算机网络实验 11		
实验学时：2	实验日期：2024.6.5	
实验目的：		

实验结果：

1、SSID “30 Munroe St”和 “linksys12”

```
, SSID="30 Munroe St"
, SSID="linksys12"
, SSID="30 Munroe St"
, SSID="linksys12"
, SSID="30 Munroe St"
, SSID="30 Munroe St"
, SSID="30 Munroe St"
, SSID="30 Munroe St"
, SSID="linksys12"
```

2、时间间隔为 0.102400 秒

```
    [FCS Status: Unverified]
    [WLAN Flags: ........C]
▾ IEEE 802.11 Wireless Management
  ▾ Fixed parameters (12 bytes)
      Timestamp: 174319001986
      Beacon Interval: 0.102400 [Seconds]
    ▸ Capabilities Information: 0x0601
  ▾ Tagged parameters (119 bytes)
```

3、源 mac 地址为 00:16:b6:f7:1d:51

```
Receiver address: Broadcast (ff:ff:ff:ff:ff:ff)
Destination address: Broadcast (ff:ff:ff:ff:ff:ff)
Transmitter address: CiscoLinksys_f7:1d:51 (00:16:b6:f7:1d:51)
Source address: CiscoLinksys_f7:1d:51 (00:16:b6:f7:1d:51)
BSS Id: CiscoLinksys_f7:1d:51 (00:16:b6:f7:1d:51)
.... .... .... 0000 = Fragment number: 0
```

4、目的 mac 地址为 ff:ff:ff:ff:ff:ff

```
    .000 0000 0000 0000 = Duration: 0 microseconds
    Receiver address: Broadcast (ff:ff:ff:ff:ff:ff)
    Destination address: Broadcast (ff:ff:ff:ff:ff:ff)
    Transmitter address: CiscoLinksys_f7:1d:51 (00:16:b6:f7:1d:51)
    Source address: CiscoLinksys f7:1d:51 (00:16:b6:f7:1d:51)
```

5、BSS Id 为 00:16:b6:f7:1d:51

```
Transmitter address: CiscoLinksys_f7:1d:51 (00:16:b6:f7:1d:51)
Source address: CiscoLinksys_f7:1d:51 (00:16:b6:f7:1d:51)
BSS Id: CiscoLinksys f7:1d:51 (00:16:b6:f7:1d:51)
.... .... .... 0000 = Fragment number: 0
```

6、

4 种速率为 1（B），2（B），5.5（B），11（B） Mbit/sec

8 种拓展速率为 6（B），9，12（B），18，24（B），36，48，54 Mbit/sec

```
> Tag: Supported Rates 1(B), 2(B), 5.5(B), 11(B), [Mbit/sec]
> Tag: Extended Supported Rates 6(B), 9, 12(B), 18, 24(B), 36, 48, 54, [Mbit/sec]
```

7、

主机 MAC 地址为：00:13:02:d1:b6:4f

第一跳路由器：00:16:b6:f4:eb:a8

AP 地址：00:16:b6:f7:1d:51

目的 IP 地址对应着第一跳路由器

```
IEEE 802.11 QoS Null function (No data), Flags: .......TC
    Type/Subtype: QoS Null function (No data) (0x002c)
  > Frame Control Field: 0xc801
    .000 0000 0010 1100 = Duration: 44 microseconds
    Receiver address: CiscoLinksys_f7:1d:51 (00:16:b6:f7:1d:51)
    Transmitter address: Intel_d1:b6:4f (00:13:02:d1:b6:4f)
    Destination address: CiscoLinksys_f7:1d:51 (00:16:b6:f7:1d:51)
    Source address: Intel_d1:b6:4f (00:13:02:d1:b6:4f)
    BSS Id: CiscoLinksys_f7:1d:51 (00:16:b6:f7:1d:51)
    STA address: Intel_d1:b6:4f (00:13:02:d1:b6:4f)
    .... .... .... 0000 = Fragment number: 0
    0101 1100 1010 .... = Sequence number: 1482
    Frame check sequence: 0x7eeda5e0 [unverified]
    [FCS Status: Unverified]
    [WLAN Flags: .......TC]
```

8、

主机 MAC 地址为：00:13:02:d1:b6:4f

第一跳路由器：00:16:b6:f4:eb:a8

AP 地址：00:16:b6:f7:1d:51

对应。

9、一个是 IP 层 DHCP release 一个是 802.11 层，Deauthentication 解除与 AP 之间的认证

```
1733 49.583615 Intel_d1:b6:4f       CiscoLinksys_f4:eb:… LLC        390 U, func=UI; SNAP, OUI 0x000000 (Officially Xerox, but 0:0:0:0
1734 49.583771                      Intel_d1:b6:4f (00:… 802.11      38 Acknowledgement, Flags=........C
1735 49.609617 Intel_d1:b6:4f       CiscoLinksys_f7:1d:… 802.11      54 Deauthentication, SN=1605, FN=0, Flags=........C
```

10、使用 wlan.fc.type_subtype == 0x000b 查找可知，共 16 条

wlan.fc.type_subtype == 0x000b

No.	Time	Source	Destination	Protocol	Lengtl	Info
1740	49.638857	Intel_d1:b6:4f	CiscoLinksys_f5:ba:…	802.11	58	Authentication, SN=1606, FN=0, Flags=.......C
1741	49.639700	Intel_d1:b6:4f	CiscoLinksys_f5:ba:…	802.11	58	Authentication, SN=1606, FN=0, Flags=....R...C
1742	49.640702	Intel_d1:b6:4f	CiscoLinksys_f5:ba:…	802.11	58	Authentication, SN=1606, FN=0, Flags=....R...C
1744	49.642315	Intel_d1:b6:4f	CiscoLinksys_f5:ba:…	802.11	58	Authentication, SN=1606, FN=0, Flags=....R...C
1746	49.645319	Intel_d1:b6:4f	CiscoLinksys_f5:ba:…	802.11	58	Authentication, SN=1606, FN=0, Flags=....R...C
1749	49.649705	Intel_d1:b6:4f	CiscoLinksys_f5:ba:…	802.11	58	Authentication, SN=1606, FN=0, Flags=....R...C
1821	53.785833	Intel_d1:b6:4f	CiscoLinksys_f5:ba:…	802.11	58	Authentication, SN=1612, FN=0, Flags=.......C
1822	53.787070	Intel_d1:b6:4f	CiscoLinksys_f5:ba:…	802.11	58	Authentication, SN=1612, FN=0, Flags=....R...C
1921	57.889232	Intel_d1:b6:4f	CiscoLinksys_f5:ba:…	802.11	58	Authentication, SN=1619, FN=0, Flags=.......C
1922	57.890325	Intel_d1:b6:4f	CiscoLinksys_f5:ba:…	802.11	58	Authentication, SN=1619, FN=0, Flags=....R...C
1923	57.891321	Intel_d1:b6:4f	CiscoLinksys_f5:ba:…	802.11	58	Authentication, SN=1619, FN=0, Flags=....R...C
1924	57.896970	Intel_d1:b6:4f	CiscoLinksys_f5:ba:…	802.11	58	Authentication, SN=1619, FN=0, Flags=....R...C
2122	62.171951	Intel_d1:b6:4f	CiscoLinksys_f5:ba:…	802.11	58	Authentication, SN=1644, FN=0, Flags=.......C
2123	62.172946	Intel_d1:b6:4f	CiscoLinksys_f5:ba:…	802.11	58	Authentication, SN=1644, FN=0, Flags=....R...C
2124	62.174070	Intel_d1:b6:4f	CiscoLinksys_f5:ba:…	802.11	58	Authentication, SN=1644, FN=0, Flags=....R...C
2156	63.168087	Intel_d1:b6:4f	CiscoLinksys_f7:1d:…	802.11	58	Authentication, SN=1647, FN=0, Flags=.......C
2158	63.169071	CiscoLinksys_f7:1d:…	Intel_d1:b6:4f	802.11	58	Authentication, SN=3726, FN=0, Flags=.......C
2160	63.169707	Intel_d1:b6:4f	CiscoLinksys_f7:1d:…	802.11	58	Authentication, SN=1647, FN=0, Flags=....R...C
2164	63.170692	CiscoLinksys_f7:1d:…	Intel_d1:b6:4f	802.11	58	Authentication, SN=3727, FN=0, Flags=.......C

11、希望打开

```
Fixed parameters (6 bytes)
    Authentication Algorithm: Open System (0)
    Authentication SEQ: 0x0001
    Status code: Successful (0x0000)
```

12、没有看到回复

13、从 49 秒到 63 秒发送 Authentication

o.	Time	Source	Destination	Protocol	Lengtl	Info
1740	49.638857	Intel_d1:b6:4f	CiscoLinksys_f5:ba:…	802.11	58	Authentication, SN=1606, FN=0, Flags=
1741	49.639700	Intel_d1:b6:4f	CiscoLinksys_f5:ba:…	802.11	58	Authentication, SN=1606, FN=0, Flags=
1742	49.640702	Intel_d1:b6:4f	CiscoLinksys_f5:ba:…	802.11	58	Authentication, SN=1606, FN=0, Flags=
1744	49.642315	Intel_d1:b6:4f	CiscoLinksys_f5:ba:…	802.11	58	Authentication, SN=1606, FN=0, Flags=
1746	49.645319	Intel_d1:b6:4f	CiscoLinksys_f5:ba:…	802.11	58	Authentication, SN=1606, FN=0, Flags=
1749	49.649705	Intel_d1:b6:4f	CiscoLinksys_f5:ba:…	802.11	58	Authentication, SN=1606, FN=0, Flags=
1821	53.785833	Intel_d1:b6:4f	CiscoLinksys_f5:ba:…	802.11	58	Authentication, SN=1612, FN=0, Flags=
1822	53.787070	Intel_d1:b6:4f	CiscoLinksys_f5:ba:…	802.11	58	Authentication, SN=1612, FN=0, Flags=
1921	57.889232	Intel_d1:b6:4f	CiscoLinksys_f5:ba:…	802.11	58	Authentication, SN=1619, FN=0, Flags=
1922	57.890325	Intel_d1:b6:4f	CiscoLinksys_f5:ba:…	802.11	58	Authentication, SN=1619, FN=0, Flags=
1923	57.891321	Intel_d1:b6:4f	CiscoLinksys_f5:ba:…	802.11	58	Authentication, SN=1619, FN=0, Flags=
1924	57.896970	Intel_d1:b6:4f	CiscoLinksys_f5:ba:…	802.11	58	Authentication, SN=1619, FN=0, Flags=
2122	62.171951	Intel_d1:b6:4f	CiscoLinksys_f5:ba:…	802.11	58	Authentication, SN=1644, FN=0, Flags=
2123	62.172946	Intel_d1:b6:4f	CiscoLinksys_f5:ba:…	802.11	58	Authentication, SN=1644, FN=0, Flags=
2124	62.174070	Intel_d1:b6:4f	CiscoLinksys_f5:ba:…	802.11	58	Authentication, SN=1644, FN=0, Flags=
2156	63.168087	Intel_d1:b6:4f	CiscoLinksys_f7:1d:…	802.11	58	Authentication, SN=1647, FN=0, Flags=
2158	63.169071	CiscoLinksys_f7:1d:…	Intel_d1:b6:4f	802.11	58	Authentication, SN=3726, FN=0, Flags=
2160	63.169707	Intel_d1:b6:4f	CiscoLinksys_f7:1d:…	802.11	58	Authentication, SN=1647, FN=0, Flags=
2164	63.170692	CiscoLinksys_f7:1d:…	Intel_d1:b6:4f	802.11	58	Authentication, SN=3727, FN=0, Flags=

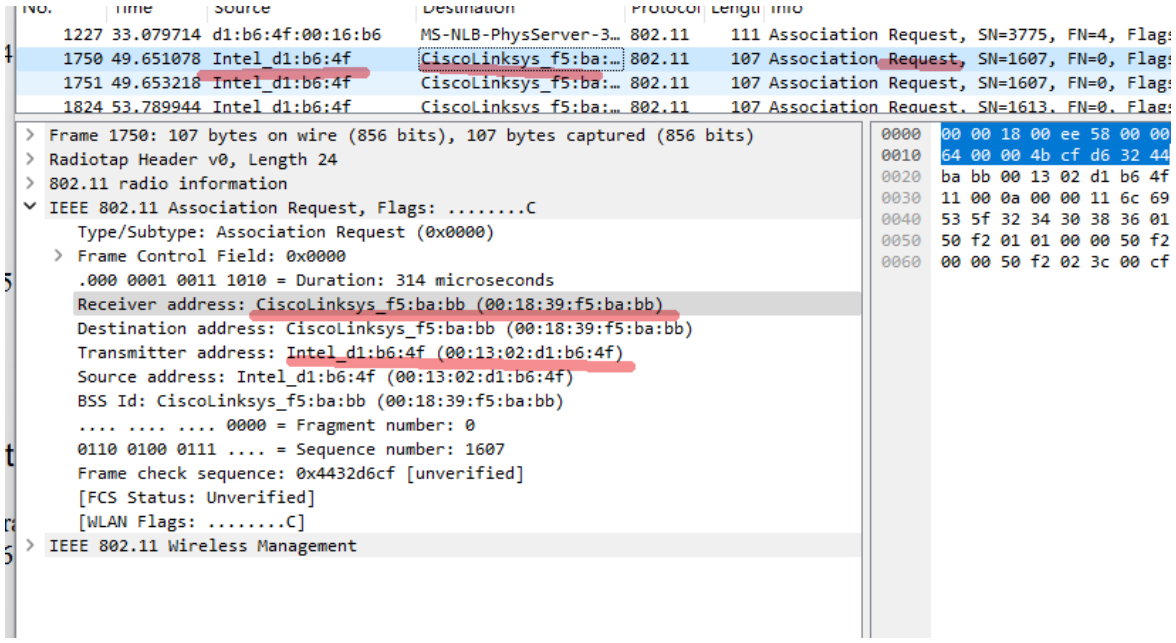
14、在 33 秒到 63 秒提出 request，在 63 秒 response

No.	Time	Source	Destination	Protocol	Lengtl	Info
1227	33.079714	d1:b6:4f:00:16:b6	MS-NLB-PhysServer-3…	802.11	111	Association Request, SN=3775, FN=4, Flags=.pm...F
1750	49.651078	Intel_d1:b6:4f	CiscoLinksys_f5:ba:…	802.11	107	Association Request, SN=1607, FN=0, Flags=.....
1751	49.653218	Intel_d1:b6:4f	CiscoLinksys_f5:ba:…	802.11	107	Association Request, SN=1607, FN=0, Flags=....R..
1824	53.789944	Intel_d1:b6:4f	CiscoLinksys_f5:ba:…	802.11	107	Association Request, SN=1613, FN=0, Flags=.....
1825	53.790943	Intel_d1:b6:4f	CiscoLinksys_f5:ba:…	802.11	107	Association Request, SN=1613, FN=0, Flags=....R..
1827	53.793568	Intel_d1:b6:4f	CiscoLinksys_f5:ba:…	802.11	107	Association Request, SN=1613, FN=0, Flags=.....
1926	57.903699	Intel_d1:b6:4f	CiscoLinksys_f5:ba:…	802.11	107	Association Request, SN=1620, FN=0, Flags=.....
1927	57.904945	Intel_d1:b6:4f	CiscoLinksys_f5:ba:…	802.11	107	Association Request, SN=1620, FN=0, Flags=....R..
1932	57.911195	Intel_d1:b6:4f	CiscoLinksys_f5:ba:…	802.11	107	Association Request, SN=1620, FN=0, Flags=....R..
1933	57.915945	Intel_d1:b6:4f	CiscoLinksys_f5:ba:…	802.11	107	Association Request, SN=1620, FN=0, Flags=....R..
1934	57.924199	Intel_d1:b6:4f	CiscoLinksys_f5:ba:…	802.11	107	Association Request, SN=1620, FN=0, Flags=....R..
1935	57.936216	Intel_d1:b6:4f	CiscoLinksys_f5:ba:…	802.11	107	Association Request, SN=1620, FN=0, Flags=....R..
1937	57.939196	Intel_d1:b6:4f	CiscoLinksys_f5:ba:…	802.11	107	Association Request, SN=1620, FN=0, Flags=.....
2126	62.176945	Intel_d1:b6:4f	CiscoLinksys_f5:ba:…	802.11	107	Association Request, SN=1645, FN=0, Flags=.....
2127	62.178194	Intel_d1:b6:4f	CiscoLinksys_f5:ba:…	802.11	107	Association Request, SN=1645, FN=0, Flags=....R..
2162	63.169910	Intel_d1:b6:4f	CiscoLinksys_f7:1d:…	802.11	89	Association Request, SN=1648, FN=0, Flags=.....
2166	63.192101	CiscoLinksys_f7:1d:…	Intel_d1:b6:4f	802.11	94	Association Response, SN=3728, FN=0, Flags=.....
2307	70.179949	CiscoLinksys_f5:ba:…	f9:ff:ff:ff:ff:ff	802.11	132	Fragmented IEEE 802.11 frame

15、以 1.0Mb/s 速率传输

```
    .0.. .... = Bad FCS: False
    0... .... = Short GI: False
Data Rate: 1.0 Mb/s
Channel frequency: 2437 [BG 6]
```

16、发送者 Mac 地址 00:13:02:d1:b6:4f，接收者 00:18:39:f5:ba:bb，BSS Id 00:18:39:f5:ba:bb。Request 帧目的是扫描所在区域中有哪些 802.11 网络； Response 帧目的是回复 Request 帧。



问题及收获：

技术问题：wlan.addr == IntelCor_d1:b6:4f 不能用于 wireshark 搜索。